

SFT V11.12 Gate 2 Addendum

Decoherence, Classical Records, and Born-Weight Consistency as Boundary-Root Resolution

Companion Addendum to SFT V11.12

Status: Standalone frozen computational scaffold addendum

Scope: Gate 2A through Gate 2X

Use: Separate companion PDF for the V11.12 archive

GATE2_ROOT_RESOLUTION_TO_CLASSICAL_RECORD_SCAFFOLD_PASS

ACTION_DERIVATION_OPEN

BORN_RULE_ASSUMPTIONS_EXPLICIT

NAMED_EXPERIMENT_COMPARISON_OPEN

Frozen scaffold draft for PDF review

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Abstract

V11.12 proposed that entanglement is shared boundary structure: a single boundary-root record serving multiple spatially separated worldlines. Gate 1 established the algebraic spine of the scalar shared-root singlet. Gate 2 asks what happens when that shared root is marked by an environment.

This addendum records the Gate 2 computational scaffold. The central result is that ordinary decoherence can be written in SFT language as *boundary-root resolution*. A shared root has an accessible coherence fraction κ . Environment marks reduce κ , increase record entropy, reduce Bell accessibility, select a pointer basis, distribute redundant records, preserve no-signaling, respect thermodynamic bookkeeping, and support a standard Born-weight scaffold under explicit symmetry, additivity, and continuity assumptions.

The scaffold does not derive the microscopic SFT action. It does not derive the Born rule from first principles. It does not claim a named calibrated experimental fit. It establishes that, once a shared-root entanglement record exists, the standard decoherence-to-classicality machinery maps cleanly onto the SFT ledger language.

One sentence version. Decoherence is the operational loss of accessible shared-root structure as environmental records resolve one boundary root into many classical boundary entries.

Reader's Guide

This addendum is intentionally separate from the main V11.12 document. The Gate 2 scaffold is large enough that placing it inside the main paper would bury the conceptual line of the argument. The recommended use is:

1. Keep the main V11.12 text focused on the shared-root proposal and Gate 1 singlet result.
2. Use this addendum as the full technical record for decoherence, classical records, and Born-weight consistency.
3. Cite this addendum wherever the main text says that root resolution reproduces the standard decoherence-to-classicality chain.

The strongest claim in this document is not that SFT has completed the microscopic action derivation. The strongest claim is that every standard constraint tested here is compatible with the boundary-root resolution dictionary, and the remaining open derivations are explicitly identified.

1. Purpose

Gate 1 answered the first question:

Can a scalar shared boundary write select the singlet record?

The Gate 1 answer was yes at the algebraic and representation level:

$$\text{scalar shared write} \Rightarrow \epsilon_{ab} \Rightarrow |S\rangle \Rightarrow E(a, b) = -a \cdot b.$$

Gate 2 asks the next question:

What happens when that shared root interacts with an environment?

The working hypothesis is:

$$\boxed{\text{decoherence} = \text{boundary-root resolution}}.$$

A shared root is not destroyed by a signal or force between distant systems. Instead, its phase coherence becomes inaccessible because the environment stores branch-distinguishing records.

Gate 2 therefore tests whether the standard decoherence and classicality chain can be written as an SFT root-ledger process without breaking known constraints.

2. Notation and Root-Resolution Dictionary

Let

$$\kappa \in [0, 1]$$

be the accessible shared-root fraction.

Then:

$$\begin{aligned} \kappa = 1 &\Rightarrow \text{fully accessible shared root,} \\ \kappa = 0 &\Rightarrow \text{fully environment-resolved root.} \end{aligned}$$

The basic SFT dictionary is:

$$\begin{aligned} \boxed{\kappa} &= \text{accessible shared-root coherence} \\ \boxed{1 - \kappa} &= \text{root-resolution fraction} \\ \boxed{S_{\text{root}}} &= \text{entropy of root-resolution ledger} \\ \boxed{\Gamma_{\text{root}}} &= \text{root-resolution rate}. \end{aligned}$$

For the two-branch singlet seed used in the Gate 2 tests:

$$C = \kappa,$$

where C is concurrence, and

$$S_{\text{CHSH}}^{\text{max}} = 2\sqrt{1 + \kappa^2}.$$

Thus Bell accessibility decays continuously from Tsirelson to the classical bound:

$$\begin{aligned} \kappa = 1 &\Rightarrow S_{\text{CHSH}}^{\text{max}} = 2\sqrt{2}, \\ \kappa = 0 &\Rightarrow S_{\text{CHSH}}^{\text{max}} = 2. \end{aligned}$$

3. Gate 2A - Per-Interaction Root Resolution

Gate 2A starts with the scalar shared-root singlet:

$$|S\rangle = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}.$$

One particle interacts with N environment markers. Each marker partially distinguishes the two branch records. Let the per-marker environment-record overlap be:

$$\mu = \langle e_{\downarrow} | e_{\uparrow} \rangle, \quad 0 \leq \mu \leq 1.$$

After N independent marks:

$$\boxed{\kappa_N = \mu^N.}$$

Tracing out the environment gives:

$$\begin{aligned} \rho_{AB}(N) = & \frac{1}{2} |\uparrow\downarrow\rangle\langle\uparrow\downarrow| + \frac{1}{2} |\downarrow\uparrow\rangle\langle\downarrow\uparrow| \\ & - \frac{\kappa_N}{2} |\uparrow\downarrow\rangle\langle\downarrow\uparrow| - \frac{\kappa_N}{2} |\downarrow\uparrow\rangle\langle\uparrow\downarrow|. \end{aligned}$$

The derived quantities are:

$$C = \kappa_N, \quad S_{\text{root}} = H_2\left(\frac{1 + \kappa_N}{2}\right), \quad S_{\text{CHSH}}^{\text{max}} = 2\sqrt{1 + \kappa_N^2}.$$

Gate 2A result: GATE2A_ROOT_RESOLUTION_DECOHERENCE_LAW_PASS

Each environment mark reduces accessible shared-root coherence by μ . Repeated marks multiply. Decoherence is therefore root-resolution per mark.

4. Gate 2B - Time-Domain Root-Resolution Rate

Gate 2A counts environment marks. Gate 2B converts marks into time.

4.1 Deterministic interaction clock

If marks arrive at rate R , then:

$$N(t) = Rt, \quad \kappa(t) = \mu^{Rt}.$$

This can be written as:

$$\kappa(t) = e^{-\Gamma_{\text{det}} t},$$

with:

$$\boxed{\Gamma_{\text{det}} = -R \ln \mu.}$$

4.2 Poisson interaction arrivals

If marks arrive as a Poisson process:

$$N \sim \text{Poisson}(Rt),$$

then:

$$\langle \kappa(t) \rangle = \mathbb{E}[\mu^N] = e^{Rt(\mu-1)}.$$

Thus:

$$\boxed{\Gamma_{\text{pois}} = R(1 - \mu).}$$

4.3 Multiple independent channels

For multiple environmental channels:

$$\boxed{\Gamma_{\text{total}} = \sum_i R_i(1 - \mu_i)}, \quad \boxed{\kappa(t) = e^{-\Gamma_{\text{total}}t}}.$$

Gate 2B result: GATE2B_ROOT_RESOLUTION_RATE_LAW_PASS

5. Gate 2C - Joint R - μ Channel Law

Gate 2C joins encounter rate and per-mark distinguishability.

For an environmental marker channel:

$$R_i = n_i \sigma_i v_i,$$

where n_i is marker density, σ_i is effective interaction cross-section, and v_i is relative speed.

The per-mark root-survival factor is modeled as:

$$\mu_i(\Delta x) = \exp \left[-\frac{1}{2} \left(\frac{\Delta x}{\ell_i} \right)^2 \right],$$

where Δx is branch separation and ℓ_i is marker resolution length.

The channel rate is:

$$\boxed{\Gamma_i(\Delta x) = R_i[1 - \mu_i(\Delta x)]}.$$

Equivalently:

$$\boxed{\Gamma_i(\Delta x) = n_i \sigma_i v_i \left[1 - \exp \left(-\frac{1}{2} (\Delta x / \ell_i)^2 \right) \right]}.$$

For $\Delta x \ll \ell_i$:

$$\frac{\Gamma_i}{R_i} \approx \frac{1}{2} \left(\frac{\Delta x}{\ell_i} \right)^2.$$

For $\Delta x \gg \ell_i$:

$$\mu_i \rightarrow 0, \quad \Gamma_i \rightarrow R_i.$$

Gate 2C result: GATE2C_JOINT_R_MU_CHANNEL_LAW_PASS

6. Gate 2D - Gaussian Marker Micro-Bridge

Gate 2D derives the overlap factor μ from explicit Gaussian environment-marker records.

Let two branch records displace a Gaussian marker by Δx . The two pointer records are centered at:

$$q = +\frac{\Delta x}{2}, \quad q = -\frac{\Delta x}{2},$$

with Gaussian width σ_q :

$$\psi_+(q) \propto \exp\left[-\frac{(q - \Delta x/2)^2}{4\sigma_q^2}\right],$$

$$\psi_-(q) \propto \exp\left[-\frac{(q + \Delta x/2)^2}{4\sigma_q^2}\right].$$

Their overlap is:

$$\mu = \langle \psi_- | \psi_+ \rangle = \exp\left[-\frac{\Delta x^2}{8\sigma_q^2}\right].$$

This matches the Gate 2C form:

$$\mu = \exp\left[-\frac{1}{2}\left(\frac{\Delta x}{\ell}\right)^2\right]$$

with:

$$\ell = 2\sigma_q.$$

The rate law becomes:

$$\Gamma = n\sigma_{\text{cross}}v \left[1 - \exp\left(-\frac{\Delta x^2}{8\sigma_q^2}\right)\right].$$

Gate 2D result: GATE2D_GAUSSIAN_MARKER_MICROBRIDGE_PASS

7. Gate 2E - σ_q Marker-Scale Audit

Gate 2E asks what can set or constrain the Gaussian marker width σ_q .

The first audited candidate is an uncertainty-limited thermal marker. For one-dimensional thermal momentum spread:

$$\sigma_p = \sqrt{mk_B T},$$

the minimum Gaussian pointer width is:

$$\sigma_{q,\text{min}} = \frac{\hbar}{2\sqrt{mk_B T}}.$$

The thermal de Broglie wavelength is:

$$\lambda_{\text{th}} = \frac{h}{\sqrt{2\pi mk_B T}}.$$

The audited relation is:

$$\sigma_{q,\text{min}} = \frac{\lambda_{\text{th}}}{2\sqrt{2\pi}}.$$

The second audited candidate is a scattering-resolution proxy:

$$\sigma_{q,\text{scatter}} = \frac{\lambda_{\text{probe}}}{2\pi}.$$

For gas-like markers:

$$n = \frac{P}{k_B T}, \quad v_{\text{mean}} = \sqrt{\frac{8k_B T}{\pi m}}, \quad R = n \sigma_{\text{cross}} v_{\text{mean}}.$$

Gate 2E result: GATE2E_SIGMAQ_SCALE_AUDIT_PASS

Gate 2E does not derive σ_q from SFT. It gives a disciplined candidate-scale dictionary.

8. Gate 2F - Kinetic Channel Parameter Bridge

Gate 2F supplies n , σ_{cross} , and v from a simple kinetic gas model.

The channel ingredients are:

$$\boxed{n = \frac{P}{k_B T}}, \quad \boxed{v_{\text{mean}} = \sqrt{\frac{8k_B T}{\pi m_{\text{gas}}}}}, \quad \boxed{\sigma_{\text{cross}} = \pi(r_{\text{system}} + r_{\text{gas}})^2}.$$

Thus:

$$\boxed{R = \frac{P}{k_B T} \pi(r_{\text{system}} + r_{\text{gas}})^2 \sqrt{\frac{8k_B T}{\pi m_{\text{gas}}}}}.$$

Combined with the Gaussian marker law:

$$\boxed{\Gamma = R \left[1 - \exp\left(-\frac{\Delta x^2}{8\sigma_q^2}\right) \right]}.$$

Gate 2F result: GATE2F_KINETIC_CHANNEL_PARAMETER_BRIDGE_PASS

Gate 2F is a kinetic bridge, not a named experimental fit.

9. Gate 2G - Collisional Decoherence Benchmark

Gate 2G compares the root-resolution law to a Gaussian momentum-kick collisional decoherence benchmark.

For a thermal gas marker:

$$\sigma_p = \sqrt{m_{\text{gas}} k_B T}.$$

Gate 2E gives:

$$\sigma_q = \frac{\hbar}{2\sigma_p}.$$

The root-resolution overlap is:

$$\mu_{\text{root}} = \exp\left[-\frac{\Delta x^2}{8\sigma_q^2}\right].$$

Substituting $\sigma_q = \hbar/(2\sigma_p)$:

$$\mu_{\text{root}} = \exp\left[-\frac{\Delta x^2 \sigma_p^2}{2\hbar^2}\right].$$

The Gaussian momentum-kick benchmark gives:

$$\mu_{\text{kick}} = \left\langle e^{iq\Delta x/\hbar} \right\rangle = \exp\left[-\frac{\Delta x^2 \sigma_p^2}{2\hbar^2}\right].$$

Therefore:

$$\boxed{\mu_{\text{root}} = \mu_{\text{kick}}}, \quad \boxed{\Gamma_{\text{root}} = R(1 - \mu_{\text{root}}) = \Gamma_{\text{benchmark}}}.$$

Gate 2G result: GATE2G_COLLISIONAL_BENCHMARK_EQUIVALENCE_PASS

Gate 2G is a benchmark-equivalence pass. It is not a named calibrated experimental comparison.

10. Gate 2H - Lindblad/Kraus Root-Resolution Bridge

Gate 2H checks that root resolution is represented by a completely positive trace-preserving dephasing channel in the branch basis.

Let:

$$\kappa(t) = e^{-\Gamma t}.$$

A branch dephasing channel multiplies the off-diagonal terms by κ :

$$\rho_{01}(t) = \kappa(t)\rho_{01}(0).$$

Equivalently, a phase-flip Kraus representation with:

$$p = \frac{1 - \kappa}{2}$$

reproduces the same root-resolution state.

The checks are:

$$\begin{aligned} \text{Tr}(\rho) &= 1, & \rho &\geq 0, & C &= \kappa, \\ S_{\text{CHSH}}^{\text{max}} &= 2\sqrt{1 + \kappa^2}, \end{aligned}$$

and no-signaling marginals are preserved.

Gate 2H result: GATE2H_LINDBLAD_KRAUS_ROOT_RESOLUTION_BRIDGE_PASS

11. Gate 2I - Pointer-Basis Stability

Gate 2I tests whether the monitored basis becomes stable.

For a pointer-basis dephasing channel:

$$\rho_{01}(t) = \kappa \rho_{01}(0),$$

while:

$$\rho_{00}(t) = \rho_{00}(0), \quad \rho_{11}(t) = \rho_{11}(0).$$

Pointer states are fixed points:

$$|0\rangle \rightarrow |0\rangle, \quad |1\rangle \rightarrow |1\rangle.$$

Superpositions lose coherence and gain entropy:

$$\alpha|0\rangle + \beta|1\rangle \rightarrow |\alpha|^2|0\rangle\langle 0| + |\beta|^2|1\rangle\langle 1|$$

in the fully resolved limit.

Gate 2I result: GATE2I_POINTER_BASIS_STABILITY_PASS

12. Gate 2J - Redundant Boundary Records / Quantum Darwinism

Gate 2J models many environment fragments receiving imperfect records.

If each fragment has branch-record overlap μ , then m fragments have overlap:

$$\mu^m.$$

For equal priors, the information available in m fragments is:

$$I(S : F_m) = H_2\left(\frac{1 + \mu^m}{2}\right).$$

The global root accessibility after N fragments is:

$$\kappa_{\text{total}} = \mu^N.$$

The tested behavior is:

1. Fragment information increases with fragment count.
2. Stronger marks require fewer fragments.
3. Redundancy increases with environment size.
4. Global coherence decays as μ^N .

Gate 2J result: GATE2J_REDUNDANT_BOUNDARY_RECORDS_PASS

Classical objectivity is redundant boundary recording. The same pointer record becomes available in many environmental fragments while the original shared root becomes inaccessible.

13. Gate 2K - Born-Weight Preservation

Gate 2K tests unequal branch weights.

For a two-branch state with supplied branch weight p :

$$|\psi\rangle = \sqrt{p}|0\rangle + \sqrt{1-p}|1\rangle.$$

Root resolution gives:

$$\rho(\kappa) = \begin{pmatrix} p & \sqrt{p(1-p)}\kappa \\ \sqrt{p(1-p)}\kappa & 1-p \end{pmatrix}.$$

The populations are preserved:

$$\rho_{00} = p, \quad \rho_{11} = 1-p.$$

The off-diagonal terms decay:

$$|\rho_{01}| = \sqrt{p(1-p)}\kappa.$$

In the fully resolved limit:

$$\kappa \rightarrow 0, \quad \rho \rightarrow p|0\rangle\langle 0| + (1-p)|1\rangle\langle 1|.$$

The entropy becomes:

$$S = H_2(p).$$

Gate 2K result: GATE2K_BORN_WEIGHT_PRESERVATION_PASS

Gate 2K does not derive the Born rule. It shows that supplied amplitude weights are preserved as classical record weights under root resolution.

14. Gate 2L - Recoherence / Quantum Eraser Boundary

Gate 2L distinguishes reversible entangling from irreversible practical root resolution.

If environment records remain coherently accessible, a unitary unmarking operation can restore the root:

$$U_{\text{mark}}^{-1} U_{\text{mark}} = I.$$

Then:

$$\rho_{\text{restored}} = \rho_{\text{initial}}.$$

If the environment is traced away or lost, local system unitaries cannot restore purity because they preserve the reduced eigenvalue spectrum:

$$\rho_S \rightarrow U_S \rho_S U_S^\dagger, \quad \text{spec}(\rho_S) \text{ is unchanged.}$$

Gate 2L result: GATE2L_RECOHERENCE_QUANTUM_ERASER_BOUNDARY_PASS

Root resolution is not fundamental destruction of the wavefunction. It becomes operationally classical when environment records are inaccessible, dispersed, or redundantly stored.

15. Gate 2M - Classical-Limit Scaling

Gate 2M checks that classicality appears as a scaling limit.

Using:

$$\Gamma = n\sigma_{\text{cross}}v \left[1 - \exp\left(-\frac{\Delta x^2}{8\sigma_q^2}\right) \right],$$

the tested scalings are:

15.1 Pressure

Since $n = P/(k_B T)$:

$$\Gamma \propto P, \quad t_{1/2} \propto P^{-1}.$$

15.2 Large-system radius

For $r_{\text{system}} \gg r_{\text{gas}}$:

$$\sigma_{\text{cross}} \approx \pi r_{\text{system}}^2,$$

so:

$$\Gamma \propto r_{\text{system}}^2, \quad t_{1/2} \propto r_{\text{system}}^{-2}.$$

15.3 Soft branch separation

For $\Delta x \ll \sigma_q$:

$$1 - \mu \approx \frac{\Delta x^2}{8\sigma_q^2},$$

so:

$$\Gamma \propto \Delta x^2.$$

Gate 2M result: GATE2M_CLASSICAL_LIMIT_SCALING_PASS

Classicality is not magic. It is a scaling regime: larger systems, higher pressures, and more distinguishable branches resolve roots faster.

16. Gate 2N - Thermodynamic Ledger / Landauer Audit

Gate 2N checks that root-resolution entropy respects thermodynamic bookkeeping.

If root resolution creates S_{bits} bits of record entropy, then erasure must obey:

$$W_{\text{min}} \geq k_B T \ln 2 S_{\text{bits}}.$$

Gate 2N audits:

1. $W/(k_B T \ln 2) = S_{\text{bits}}$.
2. Unresolved roots have zero entropy.
3. Fully resolved two-branch records have entropy $H_2(p)$.

4. Entropy is maximal at $p = 1/2$.
5. Landauer cost scales linearly with temperature.
6. Redundant record erasure cost scales linearly with record count.
7. Entropy increases as κ decreases.

Gate 2N result: GATE2N_THERMODYNAMIC_LEDGER_PASS

Gate 2N does not derive Landauer's principle from SFT. It imports Landauer as a thermodynamic constraint and shows the root-resolution ledger respects it.

17. Gate 2O - Visibility / Distinguishability Complementarity

Gate 2O checks that root resolution obeys two-branch complementarity.

Let:

$$V = \kappa$$

be phase visibility, and let:

$$D = \sqrt{1 - \kappa^2}$$

be which-root distinguishability.

Then:

$$\boxed{D^2 + V^2 = 1.}$$

As environment marking increases:

$$V \downarrow, \quad D \uparrow.$$

The Bell-accessibility bound remains:

$$2 \leq S_{\text{CHSH}}^{\text{max}} \leq 2\sqrt{2}.$$

Gate 2O result: GATE2O_VISIBILITY_DISTINGUISHABILITY_COMPLEMENTARITY_PASS

Which-root information grows exactly as phase visibility is lost.

18. Gate 2P - No-Signaling and Measurement-Order Stress Test

Gate 2P protects the causal boundary of the shared-root picture.

The tests are:

1. Local dephasing on side A does not change the marginal state on side B .
2. Measuring A then B gives the same joint probabilities as measuring B then A .
3. Sequential projective probabilities match the direct joint Born probability.
4. Remote marginals are independent of local measurement axis.
5. Reduced traces are preserved.

In density-matrix language:

$$\rho_B = \text{Tr}_A(\rho_{AB})$$

must remain invariant under local trace-preserving operations on A :

$$\rho'_B = \rho_B.$$

Gate 2P result: GATE2P_NO_SIGNALING_AND_ORDER_INVARIANCE_PASS

SFT shared roots can be nonfactorizable without permitting signaling.

19. Gate 2Q - Equal-Weight Symmetry

Gate 2Q begins the Born-weight scaffold.

For M equal-amplitude branches:

$$|a_i| = \frac{1}{\sqrt{M}}.$$

A symmetry-respecting record-weight rule must assign equal weights:

$$w_i = \frac{1}{M}.$$

The equal-branch state is invariant under branch-label permutations, so no branch can receive a privileged weight.

Gate 2Q result: GATE2Q_EQUAL_WEIGHT_SYMMETRY_PASS

This is a symmetry scaffold, not a full Born-rule derivation.

20. Gate 2R - Rational Fine-Graining

Gate 2R extends the equal-weight result to rational branch weights.

For:

$$p = \frac{m}{M},$$

write a two-branch state:

$$|\psi\rangle = \sqrt{\frac{m}{M}}|A\rangle + \sqrt{\frac{M-m}{M}}|B\rangle.$$

Fine-grain branch A into m equal microbranches and branch B into $M - m$ equal microbranches. Each microbranch has weight:

$$\frac{1}{M}.$$

Thus:

$$w(A) = m \cdot \frac{1}{M} = \frac{m}{M} = \left| \sqrt{\frac{m}{M}} \right|^2.$$

Gate 2R result: GATE2R_RATIONAL_FINE_GRAINING_BORN_WEIGHTS_PASS

This uses additivity over fine-grained equal microbranches.

21. Gate 2S - Continuity Extension

Gate 2S extends rational weights to arbitrary real weights.

Any real:

$$p \in [0, 1]$$

can be approximated by rationals:

$$p_M = \frac{m}{M}.$$

If record weights are continuous in p , then:

$$\lim_{M \rightarrow \infty} p_M = p.$$

Thus:

$$w = p = |\sqrt{p}|^2.$$

Gate 2S result: GATE2S_CONTINUITY_EXTENSION_TO_REAL_WEIGHTS_PASS

Continuity is explicit. It is not derived from SFT action dynamics here.

22. Gate 2T - Coarse-Graining / Projector Additivity

Gate 2T checks that merged records have additive weights.

For a projector onto a group of microrecords:

$$P_G = \sum_{i \in G} |i\rangle\langle i|,$$

the group weight is:

$$w(G) = \text{Tr}(\rho P_G) = \sum_{i \in G} |a_i|^2.$$

This verifies that record weights are additive under coarse-graining.

Gate 2T result: GATE2T_COARSE_GRAINING_ADDITIVITY_PASS

23. Gate 2U - POVM / Naimark Consistency

Gate 2U checks that record weights are consistent across equivalent measurement descriptions.

For a POVM:

$$\{E_k\}, \quad \sum_k E_k = I,$$

the direct probability is:

$$p_k = \text{Tr}(\rho E_k).$$

For Kraus operators M_k :

$$E_k = M_k^\dagger M_k,$$

and:

$$p_k = \text{Tr}(M_k \rho M_k^\dagger).$$

Gate 2U verifies:

$$\boxed{\text{Tr}(\rho E_k) = \text{Tr}(M_k \rho M_k^\dagger)}.$$

Gate 2U result: GATE2U_POVM_NAIMARK_CONSISTENCY_PASS

24. Gate 2V - Born Exponent Stress Test

Gate 2V tests alternative branch-weight exponents.

Consider a rule:

$$w_q(A) = \frac{|a_A|^q}{|a_A|^q + |a_B|^q}.$$

For rational fine-graining:

$$p = \frac{m}{M}, \quad a_A = \sqrt{p}.$$

Fine-graining requires:

$$w(A) = p.$$

Gate 2V tests whether:

$$\frac{p^{q/2}}{p^{q/2} + (1-p)^{q/2}} = p$$

for all rational p . The only exponent in the audited family that satisfies this is:

$$\boxed{q = 2}.$$

Gate 2V result: GATE2V_BORN_EXPONENT_STRESS_PASS; Q2_SELECTED_BY_FINE_GRAINING

This does not derive the Born rule from SFT action dynamics. It shows that, given the fine-graining scaffold, squared amplitude is selected against nearby alternative exponent rules.

25. Gate 2W - Adversarial Negative-Control Audit

Gate 2W intentionally injects bad alternatives and verifies that the Gate 2 guardrails reject them.

The negative controls are:

1. $q \neq 2$ branch-weight exponent rules.
2. Non-CPTP fake maps.
3. Signaling fake maps.
4. Non-additive coarse-graining rules.
5. Entropy-decreasing fake resolution.

The expected result is not that these alternatives pass. The expected result is that they fail.

Gate 2W result: GATE2W_ADVERSARIAL_NEGATIVE_CONTROLS_PASS

The scaffold is stronger because the good channels pass and the bad controls fail.

26. Gate 2X - Synthesis Manifest

Gate 2X builds the final manifest of Gate 2:

1. Every subgate.
2. Pass/fail status.
3. Claim level.
4. Caveat.
5. Remaining open item.

The manifest verifies:

1. All listed Gate 2 subgates pass.
2. The SFT action derivation remains explicitly open.
3. Born-rule assumptions remain explicit.
4. Named experimental comparison remains open.
5. No overclaim terms appear.

Gate 2X result: GATE2X_SYNTHESIS_MANIFEST_PASS; FREEZE_READY

27. Combined Gate 2 Law

The central root-resolution law is:

$$\kappa(t) = \exp \left[-t \sum_i n_i \sigma_i v_i \left(1 - \exp \left[-\frac{\Delta x^2}{8\sigma_{q,i}^2} \right] \right) \right].$$

Equivalently, with $\ell_i = 2\sigma_{q,i}$:

$$\kappa(t) = \exp \left[-t \sum_i n_i \sigma_i v_i \left(1 - \exp \left[-\frac{1}{2} (\Delta x / \ell_i)^2 \right] \right) \right].$$

For the two-branch singlet seed:

$$C(t) = \kappa(t), \quad S_{\text{root}}(t) = H_2 \left(\frac{1 + \kappa(t)}{2} \right),$$

$$S_{\text{CHSH}}^{\text{max}}(t) = 2\sqrt{1 + \kappa(t)^2}.$$

28. Gate 2 Status Table

Gate	Theme	Result	Caveat
2A	Per-interaction root resolution	Pass	Microscopic rate derivation open
2B	Time-domain rate law	Pass	Derivation of R and μ open
2C	Joint R - μ channel law	Pass	Channel parameters supplied, not action-derived
2D	Gaussian marker overlap	Pass	σ_q dynamics open
2E	σ_q scale audit	Pass	Candidate scales, not SFT derivation
2F	Kinetic channel bridge	Pass	Not a calibrated experiment
2G	Collisional benchmark equivalence	Pass	Named experiment comparison open
2H	CPTP dephasing bridge	Pass	Action derivation open
2I	Pointer-basis stability	Pass	Basis interaction dynamics open
2J	Redundant records / Darwinism	Pass	Specific environment calibration open
2K	Born-weight preservation	Pass	Born-rule derivation open
2L	Recoherence boundary	Pass	Practical irreversibility depends on environment access
2M	Classical-limit scaling	Pass	Calibrated macroscopic comparison open
2N	Thermodynamic ledger	Pass	Landauer imported as constraint
2O	Complementarity	Pass	Standard complementarity mapped, not action-derived
2P	No-signaling / order invariance	Pass	Causal consistency check
2Q	Equal-weight symmetry	Pass	Symmetry assumption explicit
2R	Rational fine-graining	Pass	Additivity assumption explicit
2S	Continuity extension	Pass	Continuity assumption explicit
2T	Coarse-graining additivity	Pass	Record additivity consistency
2U	POVM / Naimark consistency	Pass	Measurement-model consistency
2V	Born exponent stress	Pass	Born derivation still assumptional
2W	Negative controls	Pass	Guardrail audit
2X	Synthesis manifest	Pass	Freeze-ready scaffold

29. What Gate 2 Establishes

Gate 2 establishes the following:

1. A shared root loses accessible coherence by environment marking.
2. Independent marks multiply as $\kappa_N = \mu^N$.
3. Poisson marking gives $\Gamma = R(1 - \mu)$.
4. Multiple channels add in Γ .
5. Gaussian marker records derive $\mu = \exp[-\Delta x^2 / (8\sigma_q^2)]$.
6. Kinetic gas parameters give transparent $R = n\sigma v$ inputs.
7. The root-resolution law is equivalent to a Gaussian collisional-decoherence benchmark.
8. Root resolution is a CPTP dephasing channel in the branch basis.
9. Pointer states are fixed points.
10. Superpositions decohere into classical mixtures.
11. Redundant environment fragments store classical records.
12. Root-resolution time shrinks with pressure, system size, and branch separation.
13. Entropy obeys a Landauer bookkeeping constraint.
14. Visibility and distinguishability obey $D^2 + V^2 = 1$.
15. Local root resolution cannot signal.
16. Measurement order does not affect joint probabilities.
17. Supplied branch weights are preserved as classical record weights.
18. Equal-weight symmetry, fine-graining, continuity, additivity, POVM consistency, and exponent stress select the standard squared-amplitude scaffold.
19. Bad alternatives fail the negative-control audit.
20. The full Gate 2 manifest remains disciplined about open items.

30. What Gate 2 Does Not Establish

This section should remain prominent. It is the guardrail that keeps the addendum scientifically honest.

Gate 2 does **not** derive the microscopic SFT action for root resolution.

Gate 2 does **not** derive n , σ , v , or σ_q from first-principles SFT dynamics.

Gate 2 does **not** claim a named calibrated experimental fit.

Gate 2 does **not** derive the Born rule from the SFT action.

Gate 2 does **not** prove that collapse is fundamental.

Gate 2 does **not** prove that collapse is illusory.

Gate 2 does **not** solve every interpretation problem in quantum mechanics.

Gate 2 does **not** replace quantum field theory.

Gate 2 establishes a scaffold: if SFT supplies shared roots and a microscopic root-resolution interaction, then the standard decoherence-to-classicality machinery is recovered in boundary-ledger language.

31. Open Items After Gate 2

31.1 Open Item 1 - SFT action-level root-resolution dynamics

Derive the root-resolution interaction from the SFT action, rather than inserting it as a channel model.

Target:

$$S_{\text{SFT}} \Rightarrow \Gamma_{\text{root}}.$$

31.2 Open Item 2 - Microscopic channel parameters

Derive:

$$n, \quad \sigma, \quad v, \quad \sigma_q$$

from the actual boundary-root interaction structure.

31.3 Open Item 3 - Named experimental comparison

Compare the Gate 2 law to a calibrated decoherence experiment.

Target:

$$\Gamma_{\text{predicted}} \quad \text{vs.} \quad \Gamma_{\text{observed}}.$$

31.4 Open Item 4 - Born rule from SFT dynamics

The current Gate 2 result gives a Born-weight scaffold:

$$\text{symmetry} + \text{fine-graining} + \text{additivity} + \text{continuity} \Rightarrow w = |a|^2.$$

But the deeper SFT question remains:

Can SFT derive those assumptions from boundary-root dynamics?

31.5 Open Item 5 - Integration with the main V11.12 document

Gate 2 should be integrated as the decoherence/classicality companion to the Gate 1 singlet addendum.

32. Frozen Gate 2 Claim

The frozen Gate 2 claim is:

GATE2_ROOT_RESOLUTION_TO_CLASSICAL_RECORD_SCAFFOLD_PASS

with explicit caveats:

ACTION_DERIVATION_OPEN

BORN_RULE_ASSUMPTIONS_EXPLICIT

NAMED_EXPERIMENT_COMPARISON_OPEN

A more compact label is:

GATE2_WX_CLOSURE_AUDIT_PASS; FREEZE_GATE2_ADDENDUM_READY

33. One-Line Summary

Gate 2 shows that decoherence can be written as boundary-root resolution: environment marks reduce accessible shared-root coherence, generate classical records, preserve causality, respect thermodynamic bookkeeping, and support the standard Born-weight scaffold, while leaving the SFT action-level derivation, first-principles Born-rule derivation, and named experimental comparison explicitly open.

PDF Export Note

This addendum is designed to stand alone in the V11.12 archive.

Recommended filename:

`SFT_V11_12_Gate2_Addendum_Root_Resolution_to_Classical_Records_Scaffold.pdf`

Recommended archive placement:

`SFT_V11_12_Main.pdf`

`SFT_V11_12_Gate1_Addendum.pdf`

`SFT_V11_12_Gate2_Addendum_Root_Resolution_to_Classical_Records_Scaffold.pdf`

The filename and front matter deliberately avoid implying that the Born rule or SFT action derivation is closed.